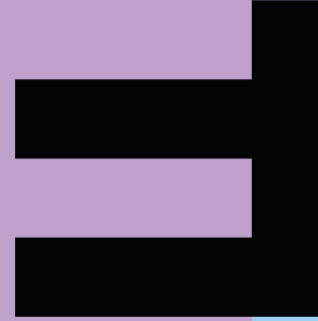


FHV

Vorarlberg University
of Applied Sciences



Digital Intro

08.04.2026

EIT & MEC Bachelor

Horatiu O. Pilsan

Boolean Algebra

- The term entropy
- The amount of information
- Unit: bit
- True/False, Yes/No, 1/0
- Boolean Algebra

$$\begin{array}{ll}\text{Konjunktion:} & y = x_1 \wedge x_2 = x_1 \cdot x_2 = x_1 x_2 \\ \text{Disjunktion:} & y = x_1 \vee x_2 = x_1 + x_2 \\ \text{Negation:} & y = \bar{x}\end{array}$$

[1], p. 623



Laws of the Boolean Algebra

Kommutatives Gesetz:			
$x_1 x_2 = x_2 x_1$	(7.1a)	$x_1 + x_2 = x_2 + x_1$	(7.1b)
Assoziatives Gesetz:			
$x_1 (x_2 x_3) = (x_1 x_2) x_3$	(7.2a)	$x_1 + (x_2 + x_3) = (x_1 + x_2) + x_3$	(7.2b)
Distributives Gesetz:			
$x_1 (x_2 + x_3) = x_1 x_2 + x_1 x_3$	(7.3a)	$x_1 + x_2 x_3 = (x_1 + x_2)(x_1 + x_3)$	(7.3b)
Absorptionsgesetz:			
$x_1 (x_1 + x_2) = x_1$	(7.4a)	$x_1 + x_1 x_2 = x_1$	(7.4b)
Tautologie:			
$xx = x$	(7.5a)	$x + x = x$	(7.5b)
Gesetz für die Negation			
$x\bar{x} = 0$	(7.6a)	$x + \bar{x} = 1$	(7.6b)
Doppelte Negation:			
$\overline{(\bar{x})} = x$	(7.7)		
De Morgans Gesetz:			
$\overline{x_1 x_2} = \bar{x}_1 + \bar{x}_2$	(7.8a)	$\overline{x_1 + x_2} = \bar{x}_1 \bar{x}_2$	(7.8b)
Operationen mit 0 und 1:			
$x \cdot 1 = x$	(7.9a)	$x + 0 = x$	(7.9b)
$x \cdot 0 = 0$	(7.10a)	$x + 1 = 1$	(7.10b)
$\bar{0} = 1$	(7.11a)	$\bar{1} = 0$	(7.11b)

[1], p. 624



Basic Functions I

a) Negation (NICHT-Verknüpfung)

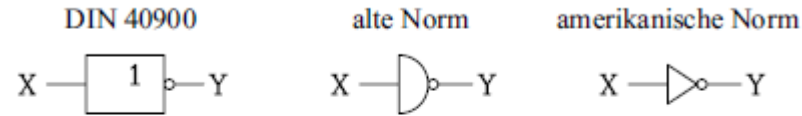


Bild 2.10: Schaltsymbole für die Negation

Wahrheitstabelle

X	Y
0	1
1	0

Funktion
 $Y = \bar{X}$

b) Konjunktion (UND-Verknüpfung)

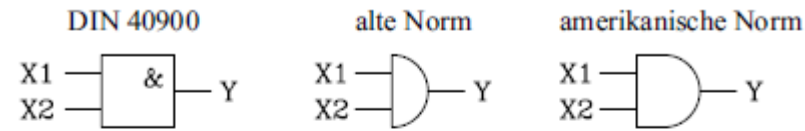


Bild 2.11: Schaltsymbole für die UND-Verknüpfung

Wahrheitstabelle

X1	X2	Y
0	0	0
0	1	0
1	0	0
1	1	1

Funktion

$$Y = X1 \cdot X2$$

Erweiterung auf n Eingangsvariablen:
 $Y = X1 \cdot X2 \cdot \dots \cdot Xn$

c) Disjunktion (ODER-Verknüpfung)

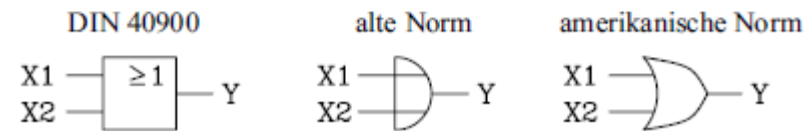


Bild 2.12: Schaltsymbole für die ODER-Verknüpfung

Wahrheitstabelle

X1	X2	Y
0	0	0
0	1	1
1	0	1
1	1	1

Funktion

$$Y = X1 \vee X2$$

Erweiterung auf n Eingangsvariablen:
 $Y = X1 \vee X2 \vee \dots \vee Xn$

[2], p. 30,31



Basic Functions II

f) Antivalenz-Verknüpfung (Exklusiv-ODER)

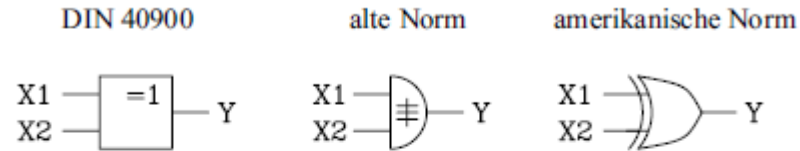


Bild 2.15: Schaltsymbole für die Antivalenz-Verknüpfung (Exklusiv-ODER)

Wahrheitstabelle		
X1	X2	Y
0	0	0
0	1	1
1	0	1
1	1	0

Funktion

$$Y = X1 \leftrightarrow X2 = \overline{X1}X2 \vee X1\overline{X2}$$

[2], p. 30,31

d) NAND-Verknüpfung

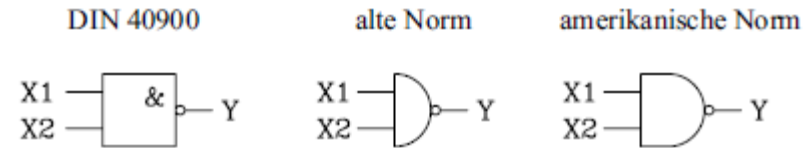


Bild 2.13: Schaltsymbole für die NAND-Verknüpfung

Wahrheitstabelle

X1	X2	Y
0	0	1
0	1	1
1	0	1
1	1	0

Funktion

$$Y = X1 \overline{X2} = \overline{X1} X2$$

Erweiterung auf n Eingangsvariablen:

$$Y = \overline{X1 X2 \dots Xn}$$

e) NOR-Verknüpfung

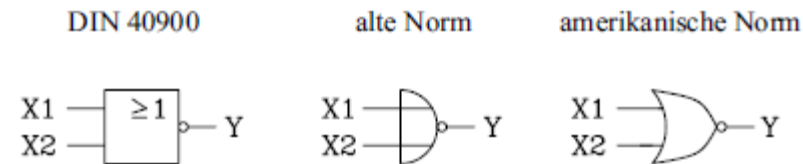


Bild 2.14: Schaltsymbole für die NOR-Verknüpfung

Wahrheitstabelle

X1	X2	Y
0	0	1
0	1	0
1	0	0
1	1	0

Funktion

$$Y = X1 \overline{X2} = \overline{X1} \vee X2$$

Erweiterung auf n Eingangsvariablen:

$$Y = \overline{X1 \vee X2 \vee \dots \vee Xn}$$



Arbitrary Functions

- An arbitrary function:

x_1	x_2	x_3	y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

x_1	x_2	x_3	y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

- Its sum of products [Disjunktive Normalform]

$$y = \overline{x_1} \cdot \overline{x_2} \cdot \overline{x_3} + \overline{x_1} \cdot x_2 \cdot x_3 + x_1 \cdot \overline{x_2} \cdot x_3$$



Transistor as a Switch

- Transistor as a Switch:

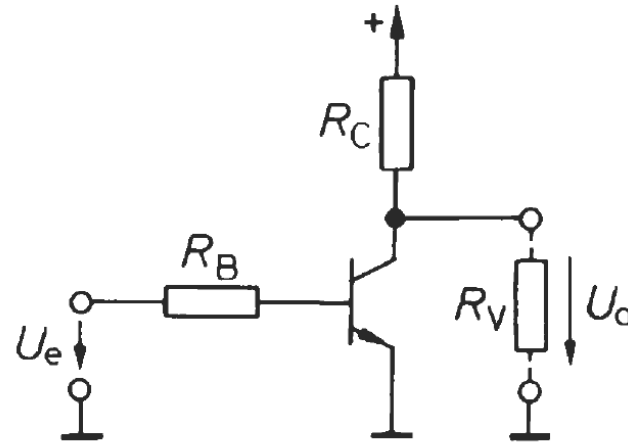
- Logic levels:

- „0“: $\leq 0.8 \text{ V}$ at the input

- „1“: $\geq 2.0 \text{ V}$ at the input

- „0“: $\leq 0.4 \text{ V}$ at the output

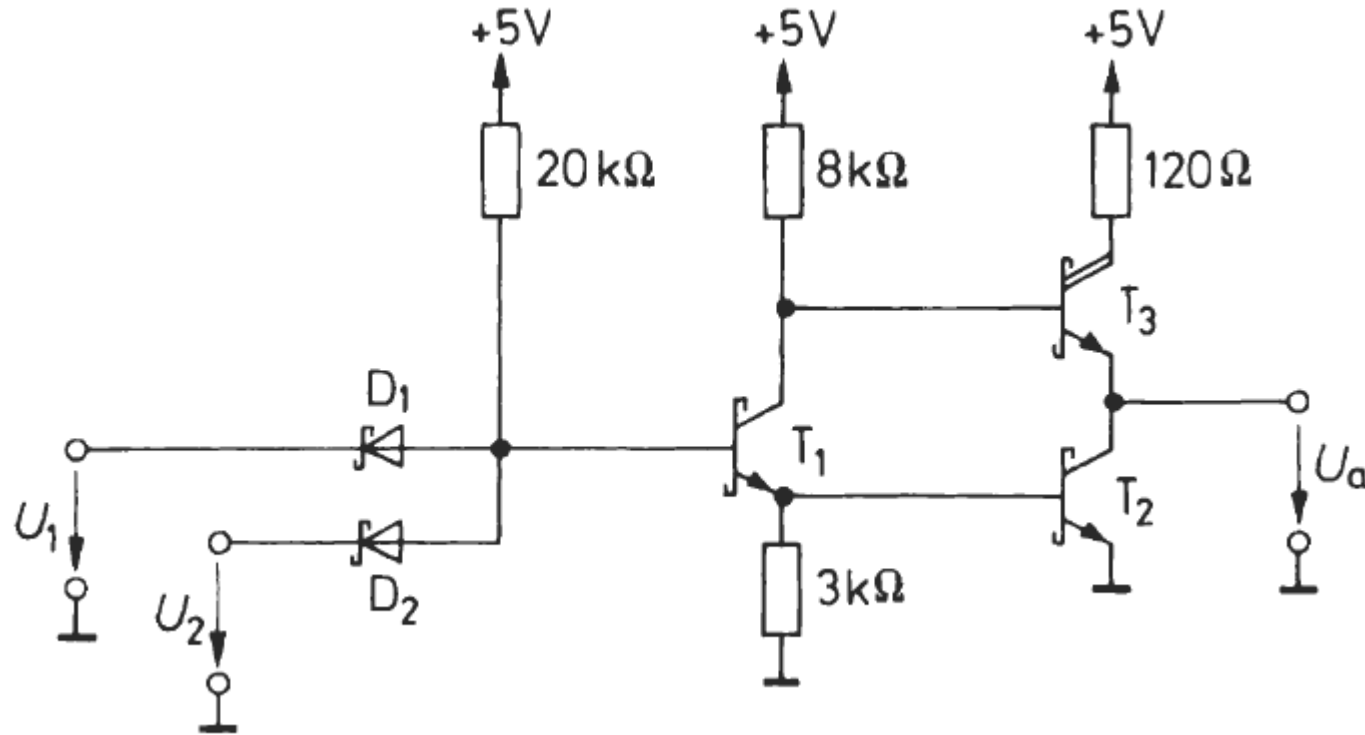
- „1“: $\geq 2.7 \text{ V}$ at the output



[1], p. 596



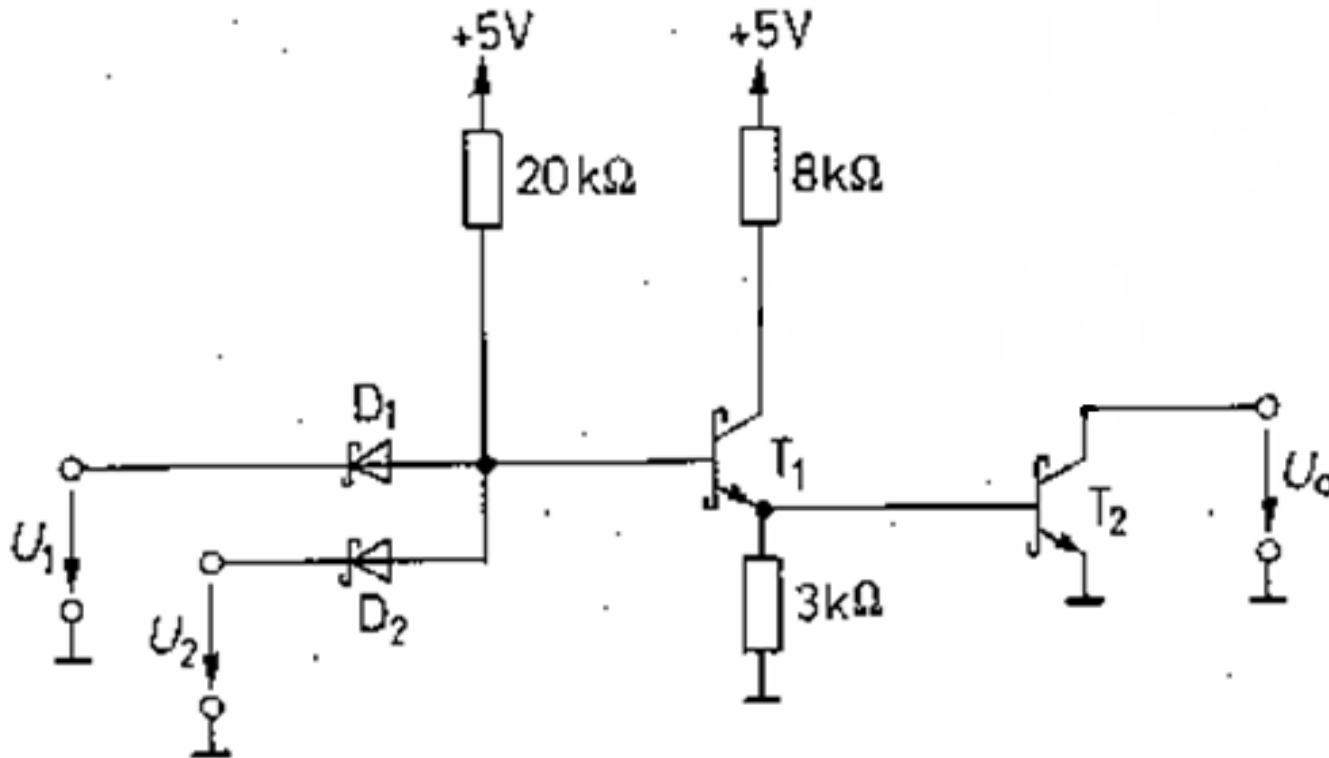
A LS TTL Gate (Bipolar Logic Example)



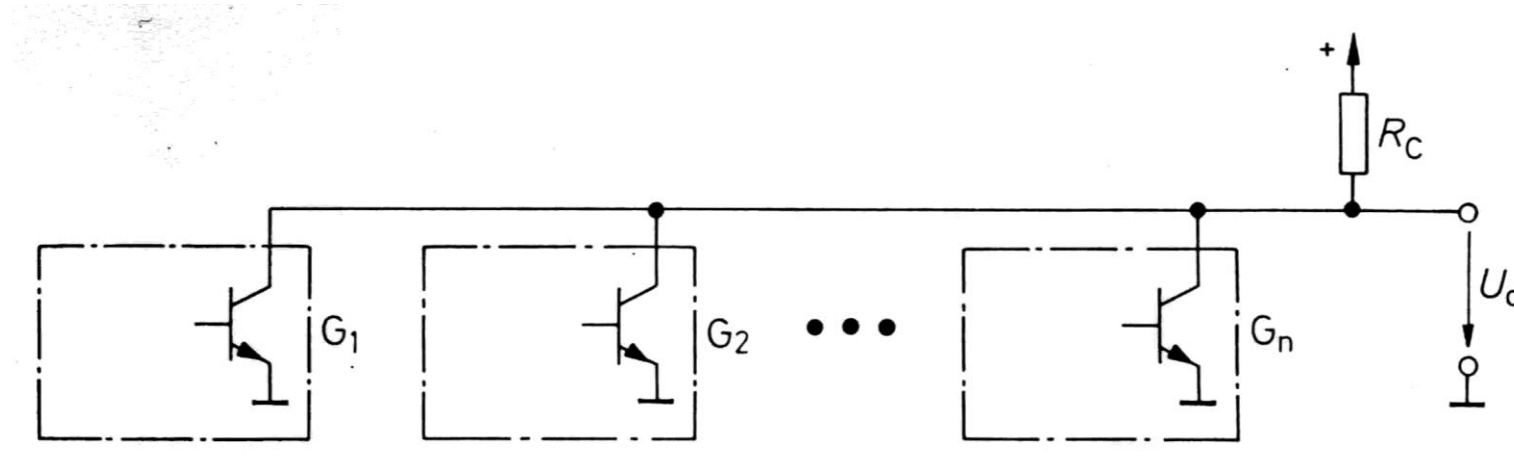
[1], p. 636



Open Collector Circuits



Application of Open Collector Circuits



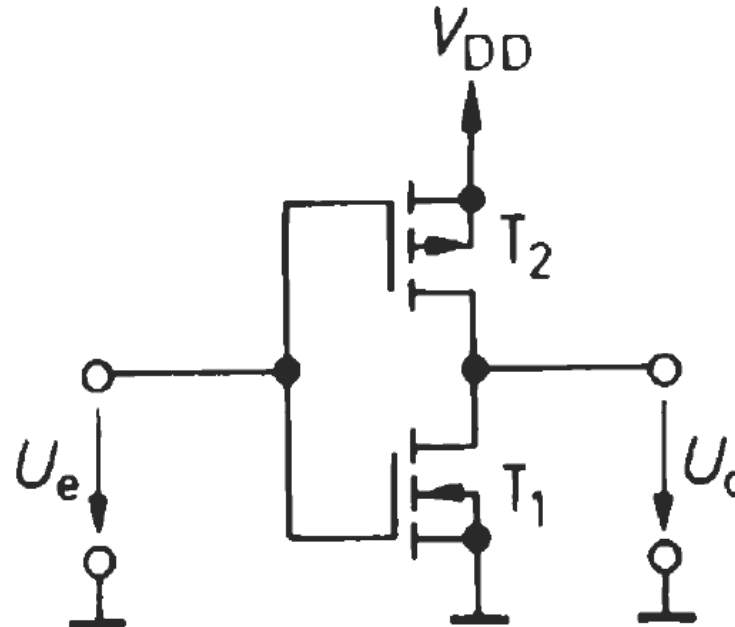
[1], p. 637



The CMOS Inverter

- The properties of a (MOS)-FET

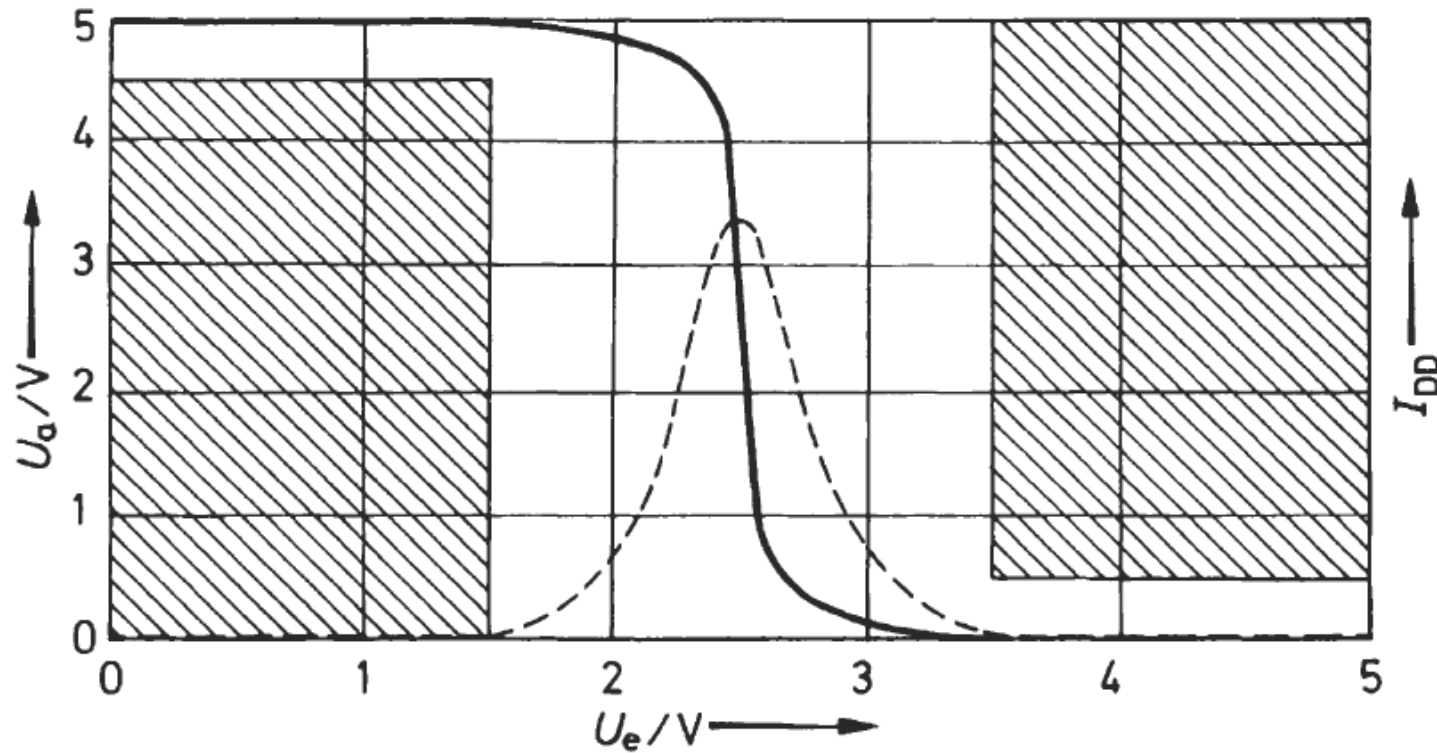
- The CMOS Inverter:



[1], p. 643



The CMOS Transition Graph



[1], p. 643



CMOS Power Consumption

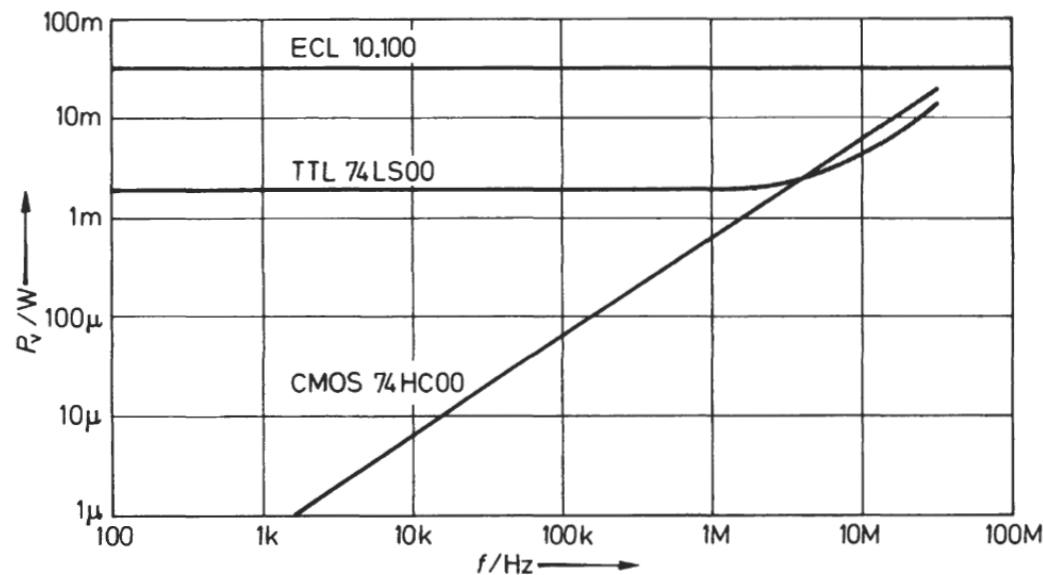
The power consumption of a CMOS logic gate is:

$$P_V = V_{DD} \cdot I = V_{DD} \cdot Q \cdot f = C_{Pv} \cdot V_{DD}^2 \cdot f$$

V_{DD} Supply voltage of the gate

I Current through the gate

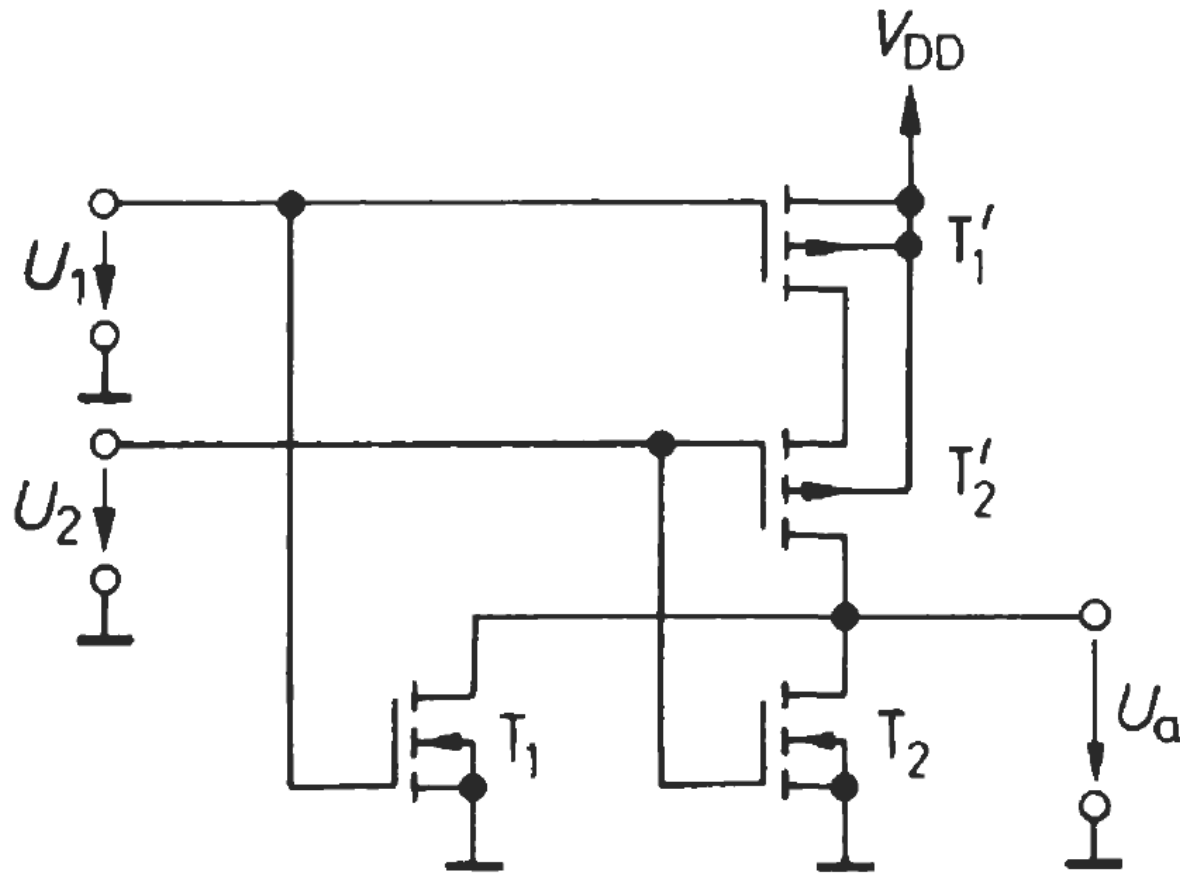
C_{Pv} fictional loss capacity



[1], p. 650



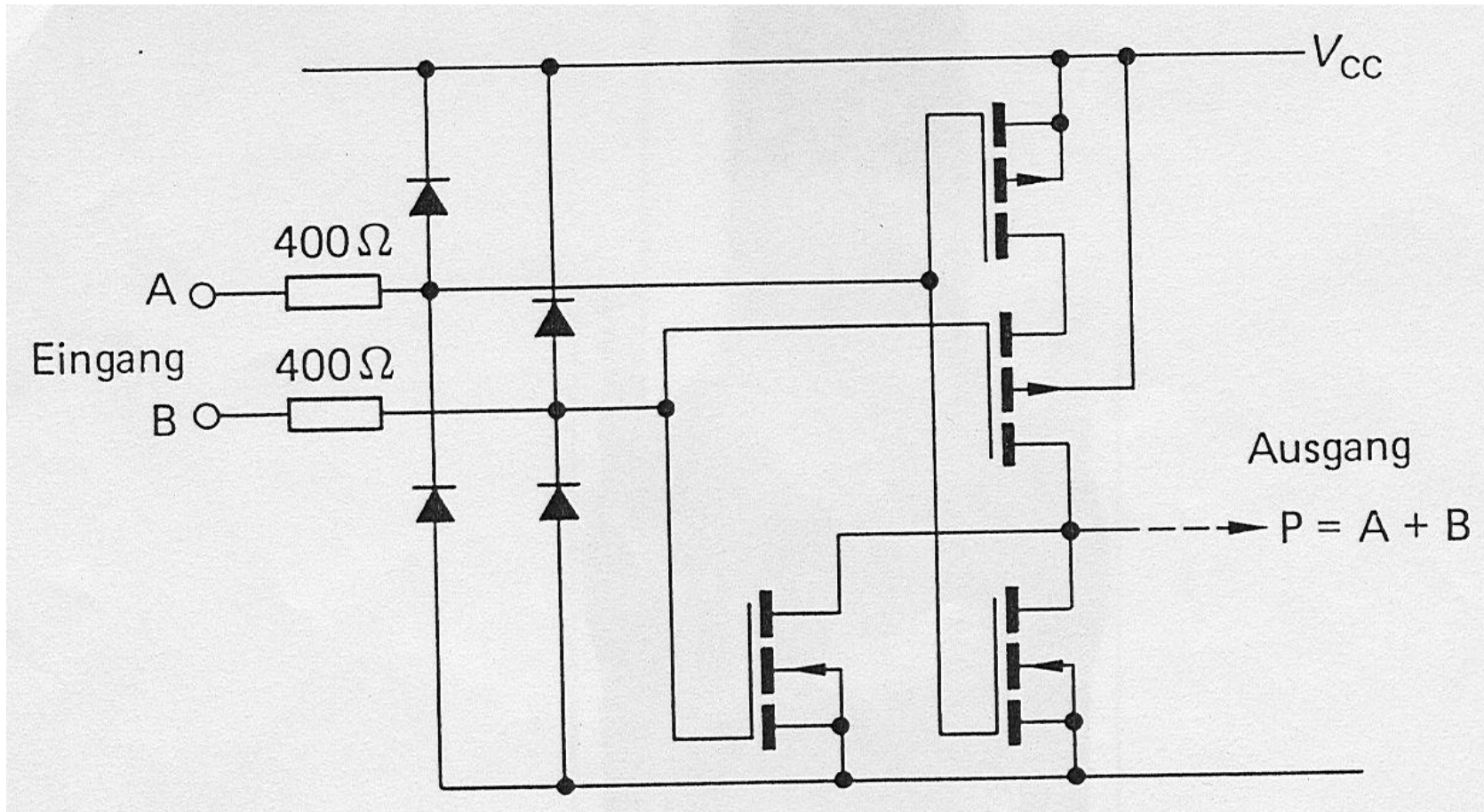
A CMOS Gate



[1], p. 645



Another CMOS Gate

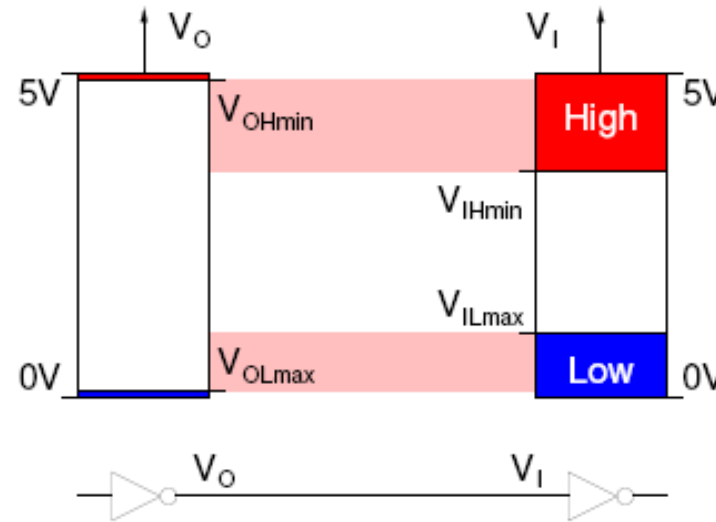


Voltage Levels

- Logic levels (supply voltage 5V):

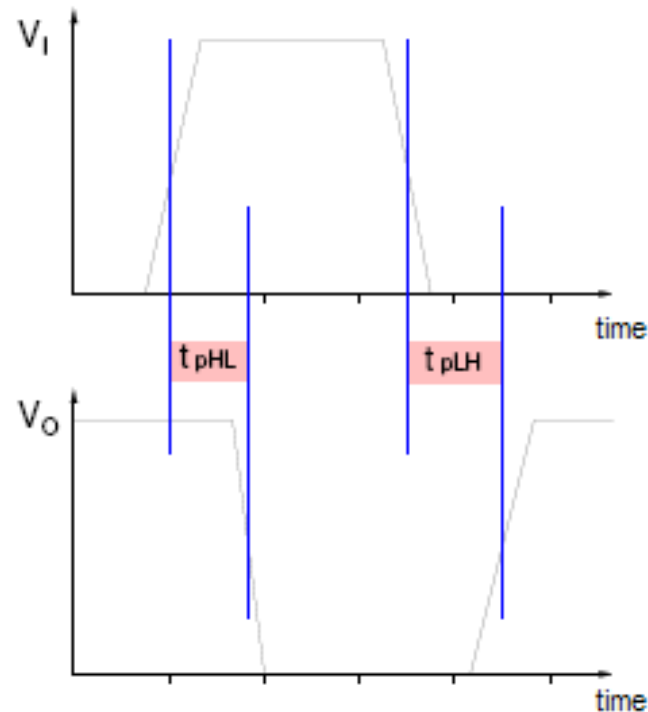
TTL In	TTL Out	CMOS In	CMOS Out
< 0.8	< 0.4	< 1.5	< 0.1
> 2.0	> 2.7	> 3.5	> 4.9

- CMOS and TTL are not compatible!
- Sufficient noise margin is essential!



Logical Gates Need Time

- Propagation through the gate takes time.
- Propagation delay times H $\rightarrow$ L and L $\rightarrow$ H are usually equal.



Logic Families (1/2)

Logic Family	Voltage Nodes, V _{CC}						Speed	Output Current		Supply Current
	1.2V	1.5V	1.8V	2.5V	3.3V	5.0V	t _{pd} max (ns)	I _{OL} (mA)	I _{OH} (mA)	I _{CC} (mA)
<u>AC</u>					○	●	6.5	24	-24	0.040
<u>AHC</u>					○	●	7.5	8	-8	0.040
<u>CBT</u>						●	0.25	-	-	0.003
<u>ABT</u>						●	3.5	64	-32	0.250
<u>ABTE</u>						●	5.2	90	-60	48
<u>74F</u>						●	6	64	-15	120
<u>BCT</u>						●	6.6	64	-15	90
<u>FB</u>						●	7	100	-3	120
<u>AS</u>						●	7.5	64	-15	143
<u>FCT</u>						●	7.5	64	-15	0.500
<u>AHCT</u>						●	7.7	8	-8	0.040
<u>ACT</u>						●	8	24	-24	0.040
<u>S</u>						●	9	64	-15	180
<u>ALS</u>						●	10	24	-15	58
<u>LS</u>						●	12	24	-15	95
<u>TTL</u>						●	22	16	-0.4	22
<u>HCT</u>						●	30	6	-6	0.080
<u>CD4000</u>						●	250	-	10	0.00025

www.ti.com



Logic Families (2/2)

Logic Family	Voltage Nodes, V _{CC}						Speed	Output Current		Supply Current
	1.2V	1.5V	1.8V	2.5V	3.3V	5.0V	t _{pd} max (ns)	I _{OL} (mA)	I _{OH} (mA)	I _{CC} (mA)
<u>AVC</u>	○	○	○	●	○		1.9	8	-8	0.040
<u>TVC</u>	○	○	○	○	○		4	-	-	-
<u>ALVC</u>			○	○	●		3	24	-24	0.040
<u>LVC</u>			○	○	●		4	24	-24	0.010
<u>CBTLV</u>				○	●		0.25	-	-	0.010
<u>ALVT</u>				○	●		2.5	64	-32	5
<u>SSTL</u>				○	●		4.1	20	-20	90
<u>LV</u>				○	●	○	14	8	-8	0.020
<u>HC</u>				○	○	●	21	6	-6	0.080
<u>LVT</u>				○	●		3.5	64	-32	0.190
<u>ALB</u>					●		2	25	-25	0.800
<u>GTLP</u>					●		TBD	100	-24	50
<u>HSTL</u>					●		5.2	24	-24	100
<u>GTL</u>					●	○	6.3	100	-32	80
<u>AC</u>					○	●	6.5	24	-24	0.040
<u>AHC</u>					○	●	7.5	8	-8	0.040

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Example of Data Sheet (1/4)

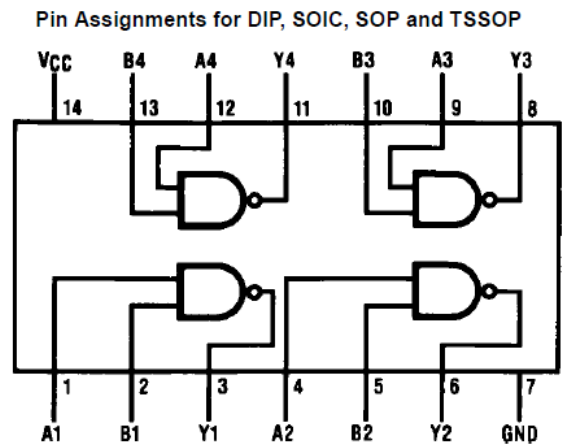
		September 1983 Revised January 2005
MM74HC00 Quad 2-Input NAND Gate		
General Description The MM74HC00 NAND gates utilize advanced silicon-gate CMOS technology to achieve operating speeds similar to LS-TTL gates with the low power consumption of standard CMOS integrated circuits. All gates have buffered outputs. All devices have high noise immunity and the ability to drive 10 LS-TTL loads. The 74HC logic family is functionally as well as pin-out compatible with the standard 74LS logic family. All inputs are protected from damage due to static discharge by internal diode clamps to V_{CC} and ground.		Features ■ Typical propagation delay: 8 ns ■ Wide power supply range: 2–6V ■ Low quiescent current: 20 μA maximum (74HC Series) ■ Low input current: 1 μA maximum ■ Fanout of 10 LS-TTL loads
Ordering Code:		
Order Number	Package Number	Package Description
MM74HC00M	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150" Narrow
MM74HC00MX_NL	M14A	Pb-Free 14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150" Narrow
MM74HC00SJ	M14D	Pb-Free 14-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
MM74HC00MTC	MTC14	14-Lead Thin Shrink Small Outline Package (TSSOP), JEDEC MO-153, 4.4mm Wide
MM74HC00MTCX_NL	MTC14	Pb-Free 14-Lead Thin Shrink Small Outline Package (TSSOP), JEDEC MO-153, 4.4mm Wide
MM74HC00N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300" Wide
MM74HC00N_NL	N14A	Pb-Free 14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300" Wide
Devices also available in Tape and Reel. Specify by appending the suffix letter "X" to the ordering code. Pb-Free package per JEDEC J-STD-020B.		

MM74HC00 Quad 2-Input NAND Gate

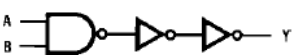


Example of Data Sheet (2/4)

Connection Diagram



Logic Diagram



MM74HC00

Absolute Maximum Ratings (Note 1)

(Note 2)

Supply Voltage (V_{CC})	-0.5 to +7.0V
DC Input Voltage (V_{IN})	-1.5 to $V_{CC}+1.5V$
DC Output Voltage (V_{OUT})	-0.5 to $V_{CC}+0.5V$
Clamp Diode Current (I_{IK}, I_{OK})	± 20 mA
DC Output Current, per pin (I_{OUT})	± 25 mA
DC V_{CC} or GND Current, per pin (I_{CC})	± 50 mA
Storage Temperature Range (T_{STG})	-65°C to +150°C
Power Dissipation (P_D)	
(Note 3)	600 mW
S.O. Package only	500 mW
Lead Temperature (T_L)	
(Soldering 10 seconds)	260°C

Recommended Operating Conditions

	Min	Max	Units
Supply Voltage (V_{CC})	2	6	V
DC Input or Output Voltage (V_{IN}, V_{OUT})	0	V_{CC}	V
Operating Temperature Range (T_A)	-40	+85	°C
Input Rise or Fall Times (t_r, t_f)			
$V_{CC} = 2V$		1000	ns
$V_{CC} = 4.5V$		500	ns
$V_{CC} = 6.0V$		400	ns

Note 1: Absolute Maximum Ratings are those values beyond which damage to the device may occur.

Note 2: Unless otherwise specified all voltages are referenced to ground.

Note 3: Power Dissipation temperature derating — plastic "N" package: -12 mW/°C from 65°C to 85°C.

Example of Data Sheet (3/4)

DC Electrical Characteristics (Note 4)

Symbol	Parameter	Conditions	V _{CC}	T _A = 25°C		T _A = -40 to 85°C		T _A = -55 to 125°C		Units
				Typ	Guaranteed Limits					
V _{IH}	Minimum HIGH Level Input Voltage		2.0V		1.5	1.5	1.5	V		
			4.5V		3.15	3.15	3.15	V		
			6.0V		4.2	4.2	4.2	V		
V _{IL}	Maximum LOW Level Input Voltage		2.0V		0.5	0.5	0.5	V		
			4.5V		1.35	1.35	1.35	V		
			6.0V		1.8	1.8	1.8	V		
V _{OH}	Minimum HIGH Level Output Voltage	V _{IN} = V _{IH} or V _{IL} I _{OUT} ≤ 20 μA	2.0V	2.0	1.9	1.9	1.9	V		
			4.5V	4.5	4.4	4.4	4.4	V		
			6.0V	6.0	5.9	5.9	5.9	V		
		V _{IN} = V _{IH} or V _{IL} I _{OUT} ≤ 4.0 mA I _{OUT} ≤ 5.2 mA	4.5V	4.2	3.98	3.84	3.7	V		
			6.0V	5.7	5.48	5.34	5.2	V		
V _{OL}	Maximum LOW Level Output Voltage	V _{IN} = V _{IH} I _{OUT} ≤ 20 μA	2.0V	0	0.1	0.1	0.1	V		
			4.5V	0	0.1	0.1	0.1	V		
			6.0V	0	0.1	0.1	0.1	V		
		V _{IN} = V _{IH} I _{OUT} ≤ 4.0 mA I _{OUT} ≤ 5.2 mA	4.5V	0.2	0.26	0.33	0.4	V		
			6.0V	0.2	0.26	0.33	0.4	V		
I _{IN}	Maximum Input Current	V _{IN} = V _{CC} or GND	6.0V		±0.1	±1.0	±1.0	μA		
I _{CC}	Maximum Quiescent Supply Current	V _{IN} = V _{CC} or GND I _{OUT} = 0 μA	6.0V		2.0	20	40	μA		

Note 4: For a power supply of 5V ±10% the worst case output voltages (V_{OH}, and V_{OL}) occur for HC at 4.5V. Thus the 4.5V values should be used when designing with this supply. Worst case V_{IH} and V_{IL} occur at V_{CC} = 5.5V and 4.5V respectively. (The V_{IH} value at 5.5V is 3.85V.) The worst case leakage current (I_{IN}, I_{CC}, and I_{OZ}) occur for CMOS at the higher voltage and so the 6.0V values should be used.



Example of Data Sheet (4/4)

MM74HC00

AC Electrical Characteristics					
$V_{CC} = 5V, T_A = 25^{\circ}C, C_L = 15\text{ pF}, t_r = t_f = 6\text{ ns}$					
Symbol	Parameter	Conditions	Typ	Guaranteed Limit	Units
t_{PHL}, t_{PLH}	Maximum Propagation Delay		8	15	ns

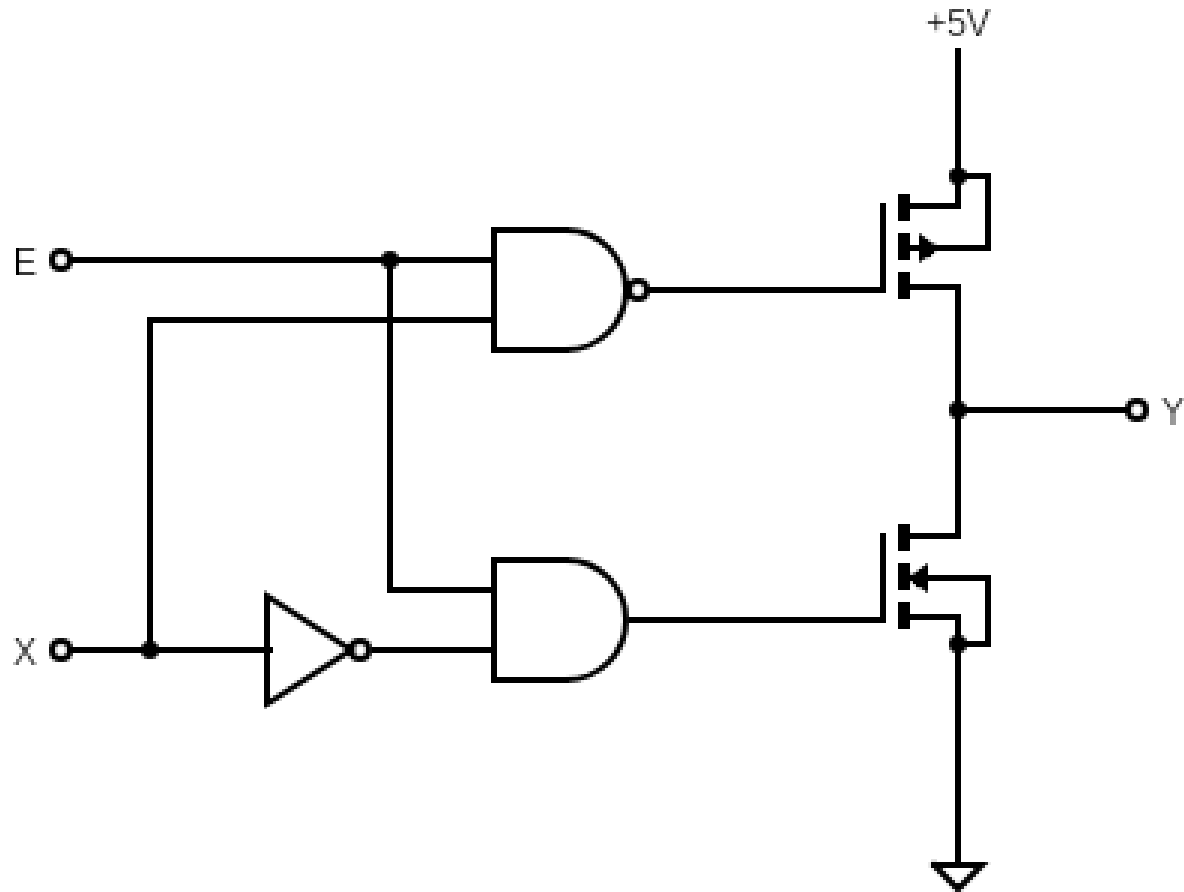
AC Electrical Characteristics								
$V_{CC} = 2.0V\text{ to }6.0V, C_L = 50\text{ pF}, t_r = t_f = 6\text{ ns}$ (unless otherwise specified)								
Symbol	Parameter	Conditions	V_{CC}	$T_A = 25^{\circ}C$		$T_A = -40\text{ to }85^{\circ}C$	$T_A = -55\text{ to }125^{\circ}C$	Units
				Typ	Guaranteed Limits			
t_{PHL}, t_{PLH}	Maximum Propagation Delay		2.0V	45	90	113	134	ns
			4.5V	9	18	23	27	ns
			6.0V	8	15	19	23	ns
t_{TLH}, t_{THL}	Maximum Output Rise and Fall Time		2.0V	30	75	95	110	ns
			4.5V	8	15	19	22	ns
			6.0V	7	13	16	19	ns
C_{PD}	Power Dissipation Capacitance (Note 5)	(per gate)		20				pF
C_{IN}	Maximum Input Capacitance			5	10	10	10	pF

Note 5: C_{PD} determines the no load dynamic power consumption, $P_D = C_{PD} V_{CC}^2 f + I_{CC} V_{CC}$, and the no load dynamic current consumption, $I_S = C_{PD} V_{CC} f + I_{CC}$.

MM74HC00

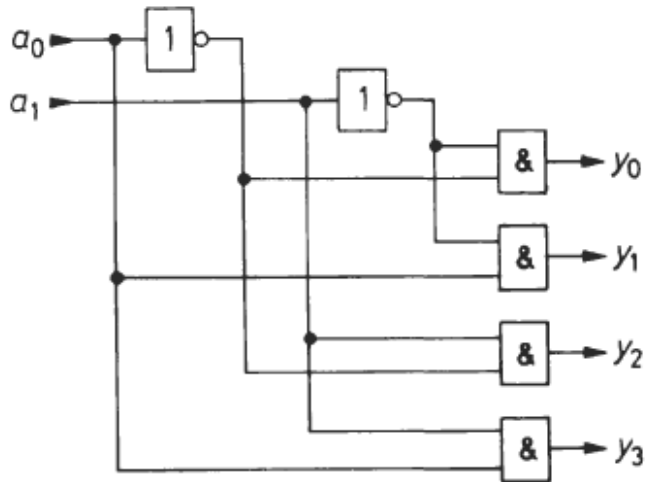


Tri State (High Impedance)



Combinational Logic I

- What is combinational logic [kombinatorische Schaltung]?
- The decoder



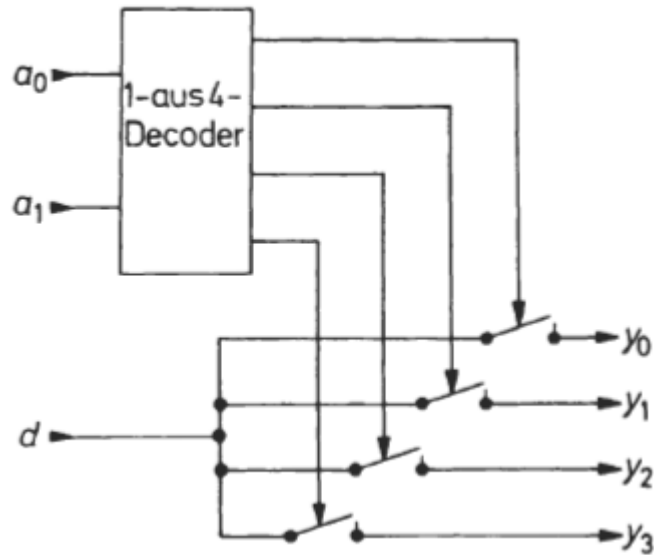
A	a_1	a_0	y_3	y_2	y_1	y_0
0	0	0	0	0	0	1
1	0	1	0	0	1	0
2	1	0	0	1	0	0
3	1	1	1	0	0	0

[1], S. 665



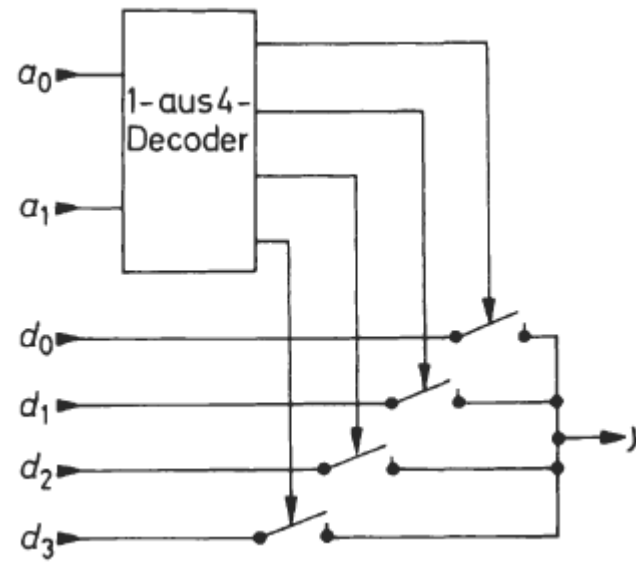
Combinational Logic II

The demultiplexer:



[1], p. 666

The multiplexer:



[1], p. 667



Binary Arithmetics

- Binary numbers
- Transformation to/from decimal numbers
- Hexadecimal numbers
- Addition of binary numbers

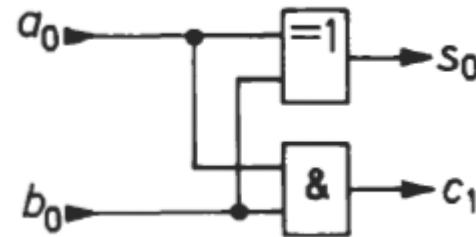
Zahlendarstellung		
dual	dez	hex
0 0 0 0	0	0
0 0 0 1	1	1
0 0 1 0	2	2
0 0 1 1	3	3
0 1 0 0	4	4
0 1 0 1	5	5
0 1 1 0	6	6
0 1 1 1	7	7
1 0 0 0	8	8
1 0 0 1	9	9
1 0 1 0	10	A
1 0 1 1	11	B
1 1 0 0	12	C
1 1 0 1	13	D
1 1 1 0	14	E
1 1 1 1	15	F



Combinational Logic III

- The (half-) adder:

a_0	b_0	s_0	c_1
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



[1], p. 673

- The full adder:

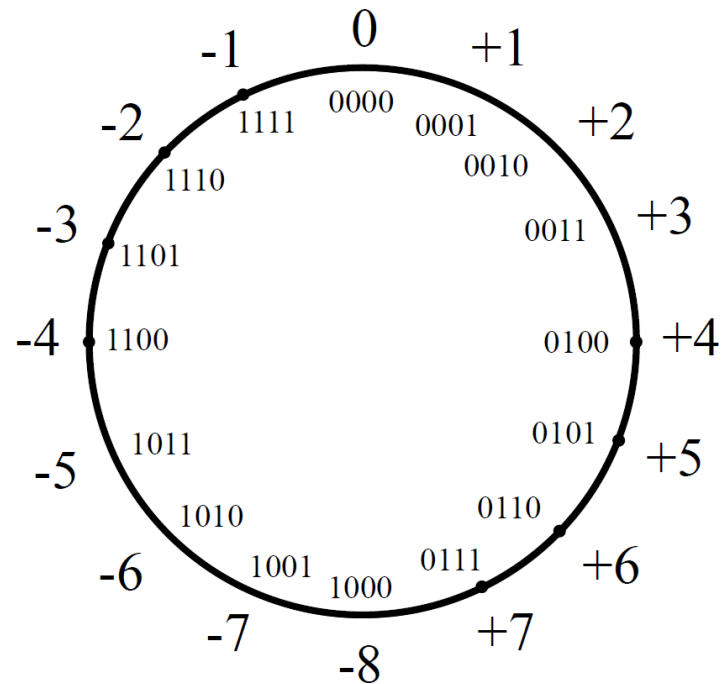
Eingang			Intern			Ausgang		Dezimal
a_i	b_i	c_i	p_i	g_i	r_i	s_i	c_{i+1}	Σ
0	0	0	0	0	0	0	0	0
0	1	0	1	0	0	1	0	1
1	0	0	1	0	0	1	0	1
1	1	0	0	1	0	0	1	2
0	0	1	0	0	0	1	0	1
0	1	1	1	0	1	0	1	2
1	0	1	1	0	1	0	1	2
1	1	1	0	1	0	1	1	3

[1], p. 674



Two's Complement

- Two's complement



- Subtraction of binary Numbers using two's complement



References

- [1] Tietze, Ulrich; Schenk, Christoph (1999): Halbleiterschaltungstechnik. 11. Auflage. Berlin: Springer
- [2] Woitowitz, Roland; Urbanski, Klaus; Gehrke, Winfried (2012): Digitaltechnik. Ein Lehr- und Übungsbuch. 6. Auflage. Heidelberg: Springer



Digital Intro

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